

Solar Energy Engineering - Photovoltaic Systems

Photovoltaic systems directly convert electromagnetic solar radiation into electrical power utilizing semiconductor materials exhibiting the photovoltaic effect. Silicon solar cells remain the dominant commercial technology.

Artificial photosynthesis aims to mimic natural biological light harvesting to produce clean hydrogen fuel and synthetic hydrocarbons through photocatalytic water splitting.

Quantum dot solar cells and perovskite multi-junction architectures offer exciting pathways toward exceeding the theoretical Shockley-Queisser limit for single-junction energy conversion.